

Teaching Statement

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1 Teaching Experience and Philosophy

Teaching has been an important part of my graduate experience, both in terms of giving back to the greater Harvard community and in deepening my own understanding of various topics. I have served as a teaching assistant (“teaching fellow”) for 5 graduate courses at Harvard across the economics and statistics departments. For *all* of these courses, I was responsible for designing *original teaching materials* for weekly (or biweekly) supplementary, lecture-based teaching sessions (“sections”).

I care quite a bit about pedagogy for statistical methods, and strive for an “intuition first” approach to statistical and econometric instruction: emphasizing the motivation for methods before delving into their formalization. I think this approach is particularly well-suited to younger students or those with less formal mathematical training, both of whom I think I have especially good theory of mind for, having started my formal mathematical and statistical training later in my academic career than many of my peers. I also think this approach applies to proof writing: leading with intuition for a high-level proof roadmap before enhancing detail and deriving technical results with full rigor. When students are struggling, I often turn to one-on-one conversation and board work, for which I think there is very little substitute when it comes to fostering deeper understanding of tough technical concepts.

Given my interdisciplinary work in machine learning methods, I also think I have a comparative advantage in instruction for state of the art advances in AI, both at an intuitive and technical level. A key input to my research involves staying on top of the latest technical work in AI/ML theory and application, and I have experience synthesizing and relaying insights from this ever-growing literature to students that come from social sciences (but not necessarily CS or statistics) backgrounds, both in teaching and in my ordinary research communications.

2 Courses Taught

1. ECON 2148, Causal Inference in Economics (for Professor Davide Viviano, Spring 2025 and 2026): A second-year elective for graduate students pursuing more advanced econometrics research. Topics covered included: selection on observables; LATE; marginal treatment effects; DAGs; mediators and surrogates; dynamic treatments; sampling and design-based uncertainty; hypothesis testing; inference and design of stratified experiments; inference with clusters and spatial dependence; multiple hypothesis testing; regression adjustment, double robustness, and augmented IPW; penalized regression; empirical Bayes methods; offline and online policy learning; DiD methods; factor models, synthetic control, and matrix completion methods; models of spillovers; and shift-share designs and recentered instruments.

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2. STAT 242/ECON 2142, Time Series (for Professor Neil Shephard, Fall 2024 and 2025):
An advanced graduate course in time series methods. Topics covered included: martingales, Brownian motion, stochastic integration; theory of prediction; linear models and projection; causality (e.g., local projection, control); hidden Markov models, filtering and smoothing; stationarity and non-stationarity; and stochastic volatility and high frequency analysis.
3. ECON 2355, Unleashing Novel Data at Scale (for Professor Melissa Dell, Spring 2021):
An elective graduate course in developing and applying state of the art AI/ML methods for empirical economics research. Topics covered included: deep learning; convolutional neural networks and image classification; object detection; active learning; generative adversarial networks; optical character recognition; dependency parsing, part of speech tagging, and named entity recognition; language models; the transformer and transformer-like architectures; information retrieval and question answering; zero-shot and few-shot learning; natural language generation and summarization; and low-resource NLP.

3 Future Courses

Given my teaching and research experiences, I would be excited to teach a core econometrics graduate sequence, as well as introductory or graduate material in asymptotic statistics. Electives of interest to me that I feel capable of teaching well include those on: state of the art econometric methods; time series statistical and causal inference; AI/ML methods in econometrics (semiparametric inference and debiased machine learning); selective inference (multiple testing, knockoffs); machine learning methods for the social sciences (including computer vision and NLP); data science for informing policy; or some other elective at the intersection of statistical methods, social science, and AI/ML.

4 Teaching Evaluations

Below are my teaching evaluation scores as well as *all* written student feedback I received. Evaluation scores displayed are responses to the question “Evaluate your Section Leader overall,” scored on a scale of 1 (unsatisfactory) to 5 (excellent).

1. ECON 2148, Causal Inference in Economics (Graduate), Teaching Fellow to Professor Davide Viviano (Spring 2025, Spring 2026)
 - Spring 2025: mean rating of 5.00 (vs. department mean of 4.42); Spring 2026: mean rating of 5.00 (vs. department mean of 4.69).
 - All student comments:
 - Jake is great, knowledgeable, and very happy to answer questions.
 - Great! I learned a lot from section and always had fun.
2. STAT 242/ECON 2142, Time Series (Graduate), Teaching Fellow to Professor Neil Shephard (Fall 2024, Fall 2025)
 - Fall 2024: mean rating of 4.67 (vs. department mean of 4.46); Fall 2025; mean rating of 5.00 (vs. department mean of 4.69).
 - All student comments:
 - Jake was great! Very helpful and approachable.

3. ECON 2355, Unleashing Novel Data at Scale (Graduate), Teaching Fellow to Professor Melissa Dell (Spring 2021)
- Mean rating of 5.0 (vs. department mean of 4.54).
 - All student comments:
 - Jake was an excellent TF! His feedback on the progress reports was incredibly helpful and he was super responsive. He really helped so much.
 - Jake’s knowledge about different CV and NLP resources has greatly helped us to improve our project
 - Deeply knowledgeable and good at communicating complex technical concepts.
 - Jacob is an incredible TF who was always accessible outside of class, eager to help, and very knowledgeable. One of the best I’ve had.